

Competitive Exam Practice 1: Foundations, Pointers & ADTs

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type; TF = True/False. C is the language.

Q1 [*GATE CS 2001 Q2.18, verified — GATEOverflow V2 p.293*] [MCQ, 2 marks] Consider the following three C functions:

```
/* P1 */
int *g(void) {
    int x = 10;
    return (&x);
}

/* P2 */
int *g(void) {
    int *px;
    *px = 10;
    return px;
}

/* P3 */
int *g(void) {
    int *px;
    px = (int *) malloc(sizeof(int));
    *px = 10;
    return px;
}
```

Which of the above three functions are likely to cause problems with pointers?

- (A) Only P3
- (B) Only P1 and P3
- (C) Only P1 and P2
- (D) P1, P2 and P3

Q2 [*GATE CS 2014 Set 1 Q10, verified — GATEOverflow V2 p.295*] [MCQ, 1 mark] Consider the following program in C language:

```
#include <stdio.h>
main() {
    int i;
    int *pi = &i;
    scanf("%d", pi);
    printf("%d\n", i + 5);
}
```

Which one of the following statements is TRUE?

- (A) Compilation fails.
- (B) Execution results in a run-time error.
- (C) On execution, the value printed is more than the address of variable `i`.
- (D) On execution, the value printed is more than the integer value entered.

Q3 [*GATE CS 2017 Set 1 Q13, verified — GATEOverflow V2 p.296*] **[MCQ, 1 mark]** Consider the following C code:

```
#include <stdio.h>
int *assignval(int *x, int val) {
    *x = val;
    return x;
}
void main() {
    int *x = malloc(sizeof(int));
    if (NULL == x) return;
    x = assignval(x, 0);
    if (x) {
        x = (int *)malloc(sizeof(int));
        if (NULL == x) return;
        x = assignval(x, 10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

- (A) compiler error as the return of `malloc` is not typecast appropriately.
- (B) compiler error because the comparison should be made as `x == NULL`.
- (C) compiles successfully but execution may result in dangling pointer.
- (D) compiles successfully but execution may result in memory leak.

Q4 [*GATE CS 2025 Set 1 Q24, verified — GATEOverflow V2 p.297*] **[NAT, 1 mark]** What is the output of the given C program?

```
#include <stdio.h>
void foo(int *p, int x) {
```

```

        *p = x;
    }
    int main() {
        int *z;
        int a = 20, b = 25;
        z = &a;
        foo(z, b);
        printf("%d", a);
        return 0;
    }

```

The output of the given program is _____. (Answer in integer)

Q5 [*Original, GATE-pattern*] [MCQ, 1 mark] In a running C program, where does a local variable declared inside a function live, and where does memory requested by `malloc` live, respectively?

- (A) heap; stack
- (B) stack; heap
- (C) stack; stack
- (D) heap; heap

Q6 [*Original, GATE-pattern*] [TF, 1 mark] State whether the following statement is TRUE or FALSE:

A block of memory obtained by `malloc` inside a function survives the return of that function, whereas a local variable of the same function does not.

Q7 [*Original, GATE-pattern*] [TF, 1 mark] State whether the following statement is TRUE or FALSE:

NULL is a pointer value meaning “points to nothing,” and dereferencing a NULL pointer is undefined behaviour — in practice, a crash.

Q8 [*Original, GATE-pattern*] [MCQ, 1 mark] Which of the following statements about `malloc` is TRUE?

- (A) On success `malloc` returns the number of bytes allocated.
- (B) On success `malloc` returns the address of the requested block; on failure it returns NULL.
- (C) `malloc` always returns NULL once the program has used more than half of its memory.
- (D) A pointer obtained from `malloc` must be dereferenced immediately or the program crashes.

Q9 [*Original, GATE-pattern*] [NAT, 2 marks] On a 64-bit machine where an `int` occupies 4 bytes and a pointer occupies 8 bytes (alignment: fields are aligned to their own size), how many bytes does the following structure occupy?

```

struct node {
    int data;
    struct node *next;
};

```

The number of bytes occupied is _____.

Q10 [*Original, GATE-pattern*] [**MCQ, 2 marks**] Recall Sedgewick's definition of a data type as "a set of values and a set of operations on those values," and the Week 1 **bag** ADT supporting `insert(x)`, `member(x)`, and `size()`. Which statement is TRUE?

- (A) "The values are stored in a dynamic array" is part of the bag's contract.
- (B) `insert(x)`, `member(x)`, and `size()` name the operations of the contract, which does not say whether the bag is implemented with an array or a linked list.
- (C) An ADT is one particular implementation of a data structure.
- (D) The contract must also specify the C code for every operation.